airtel

SYNC

Project Udyam Digital

Turning fragmented MSME onboarding into one connected digital-to-assisted journey–helping merchants discover, quote, verify, and connect seamlessly.



Project Udyam Digital · **Enterprise Track A** Business case : Team Kastu
Github :github.com/vipulMadan31/Airtel-Hackathon (also contains 1 page DPDP Act Note)
Demo Video :
https://drive.google.com/file/d/1LAw99r_6gE6zIL88bekeCp_QOTf2Z2ZZ/view?usp=sharing

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The Acquisition Bottleneck (Baseline Funnel)

The Current Acquisition Funnel

Funnel Progression (5,000 Inbound Leads):
1. Inbound Leads: 5,000 (100%) 2.Contacted: 3,406 (68.1%) 3. Needs Assessed: 1,974 (39.5%)
4. Quote Accepted: 861 (17.2%) 5. KYC Complete: 588 (11.8%) — *31.7% drop-off post-quote*
6. Activated Connections: 368 (7.36%)

Key Baseline Metrics:
7.36% Overall Activation Conversion
17.62 Days Mean Time to Activation (Std Dev: 4.21 days)
₹15,053.37 Current Cost per Activation (CAC)
3.01 Mean Status-Chasing Calls per Activated Customer

Primary Drop Reasons:
805: Unreachable after 3 contact attempts
531: Merchant has no time for physical sales visit
414: Price perceived too high
385: Quote took too long (179) or was too complex (206)
173: Missing GST/Udyam documents (94) or paperwork too lengthy (79)
41: Site not feasible for fibre

Notes:
"Analyzing our baseline dataset of 5,000 leads shows that only 7.36% convert to active billing connections, with onboarding taking an average of 17.6 days. Over 1,300 leads drop because merchants are unreachable or lack time for physical visits, and another 385 drop due to slow, opaque quoting. The current physical-first sales process creates friction that loses high-intent leads."

Source: Project baseline dataset (app/data/smb_leads.csv, synthetic 5,000 leads). Internal illustrative data.

Architecture Overview

Unified Digital-to-Assisted Architecture

Surface 1: Customer Web Surface
Lightweight responsive storefront built in HTML, CSS, and vanilla JavaScript
Native multilingual interface (English, Hindi, Marathi)
Browser voice typing enabled in English and Hindi

Surface 2: Conversational Channel
WhatsApp local simulator and webhook scaffold
Conversational discovery, instant quote review, and automated tracking alerts
Disclaimer: Simulated local scaffold; not connected to live Meta WhatsApp Business API.

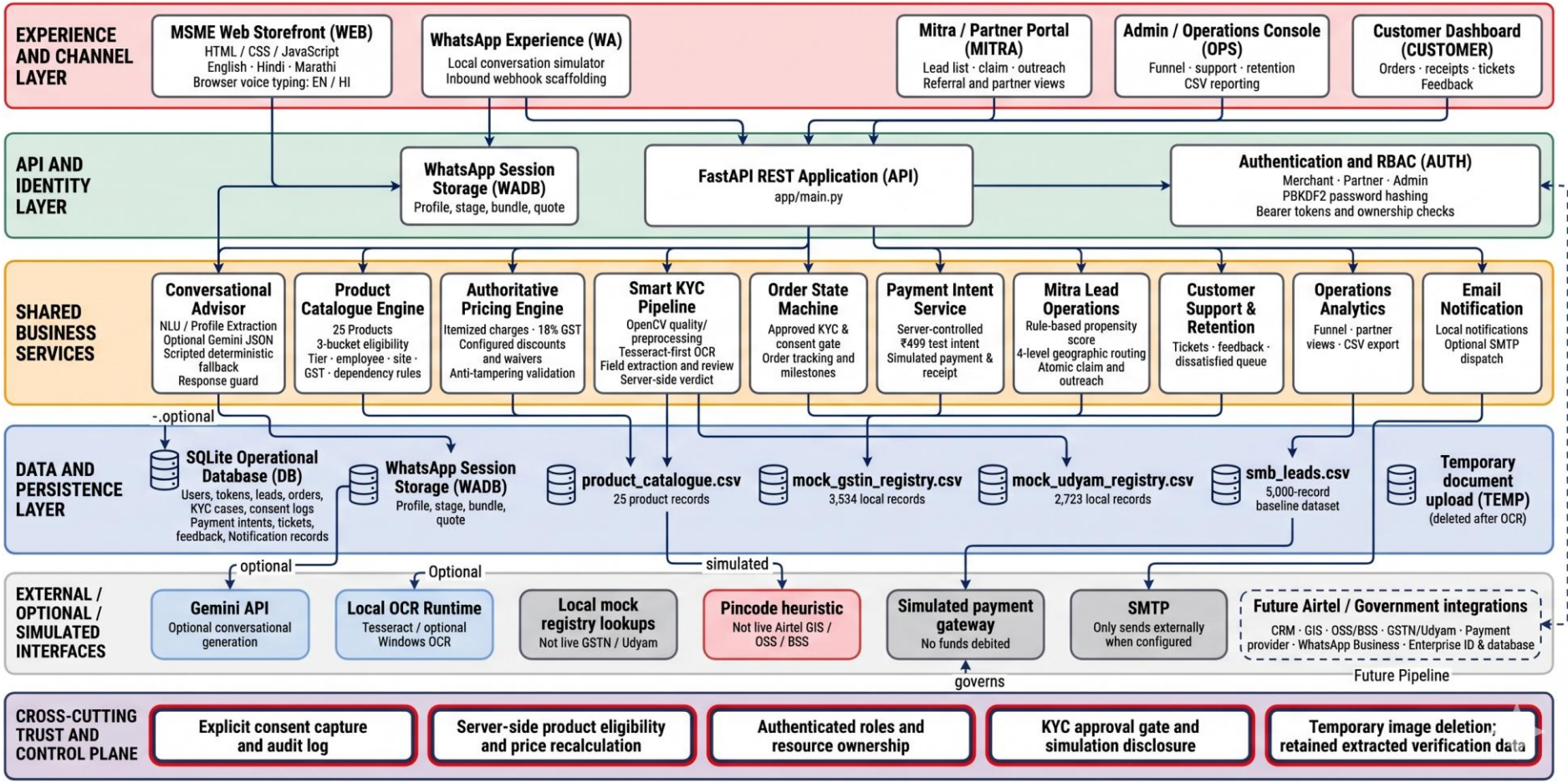
Surface 3: Mitra Partner & Ops Portals
Dedicated partner portal for assisted sign-ups and geographic lead claiming
Central operations console for pipeline tracking and retention intervention
Real-time CSV exports and funnel status queues

Core Engine:
Unified Python/FastAPI business layer with local SQLite persistence

Notes:
"We developed a modular architecture across three primary surfaces: a lightweight, self-serve web application, a simulated WhatsApp conversational workflow, and dedicated consoles for our Mitra channel and central operations. All three feed into a unified FastAPI backend to ensure continuous state tracking when moving between digital self-service and human-assisted sales."

PROJECT UDYAM DIGITAL | SOLUTION ARCHITECTURE

PROTOTYPE ONLY



Needs Assessment

Intelligent Discovery & Dynamic Eligibility

Smart Advisor Architecture:

Natural Language Understanding (NLU) extracts merchant profile, site counts, and intent
Optional Gemini integration generates structured conversational JSON
Strict server-side **Response Guard** prevents price hallucination or unauthorized bundling
Deterministic scripted fallback guarantees continuous flow if LLM connectivity is lost
Note: Designed as a bounded conversational interface, not an autonomous tool-calling agent.

Dynamic Eligibility Engine:

Classifies a 25-product enterprise catalogue across 3 distinct buckets: Base Plans, Eligible Add-ons, and Ineligible/Locked Services
Real-time eligibility logic evaluates:
City Tier & Pincode Feasibility *(evaluated via local heuristic simulation)*
Employee headcount and branch site volume
Tax registration status (GST registered vs. non-GST micro-enterprise)
Minimum contract durations and cross-product technical dependencies

Speaker Notes:

"Our discovery engine lets merchants express requirements conversationally in their local language. Gemini powers the conversational extraction, but it operates inside strict boundaries: a response guard prevents unauthorized quotes, and an automated scripted fallback prevents stalled sessions. Eligibility is evaluated across our 25-product catalogue using deterministic rules for location, headcount, and GST status."

Quote Generation

Server-Authoritative Transparent Pricing

Quotation Architecture:

Server-Side Repricing: All calculations and discount rules run on the FastAPI backend to block client-side tampering
Interactive Customization: Merchants add or remove eligible add-ons with immediate price recalculation
Configured Rules: Promotional setup fee waivers apply automatically based on catalogue rules

Validated End-to-End Test Scenario:

Biz Fibre 100 (Catalogue ID: P01): ₹999.00 /month
Cloud POS & Billing (Catalogue ID: P19): ₹599.00 /month
Monthly Subtotal: ₹1,598.00
Applicable GST (18.0%): ₹287.64
Total Monthly Commitment: ₹1,885.64 /month
One-Time Setup Fee: ₹0.00 *(₹2,500 standard catalogue fee waived via prototype promo)*
Booking Token Required: ₹499.00

Notes:

"To eliminate pricing opacity, quotes are calculated entirely server-side. In our validated test scenario, a merchant configuring Biz Fibre 100 and Cloud POS receives an authoritative monthly subtotal of ₹1,598, with 18% GST calculated to ₹287.64, yielding an exact monthly bill of ₹1,885.64. The ₹2,500 setup fee is automatically waived by the rules engine, and an advance booking token of ₹499 locks the order.

Values reflect prototype catalogue configuration rules; not live commercial Airtel Business rate cards.

KYC Workflow

Dual-Track Verification & Consent Framework

Document Processing Pipeline:
Client-side capture with automated OpenCV blur and quality checks
Local Tesseract OCR processing with automatic Windows OCR fallback
User-editable interface allows merchants to correct any OCR extraction variance directly

Dual Registry Validation Paths (Simulated):
Track A (GST Enterprises): Automated validation of GSTIN against mock commercial tax registries
Track B (Micro Enterprises): Tailored flow validating Udyam Registration Numbers (URN) for non-GST merchants
Direct address and entity name consistency matching across documents

Data Privacy & Consent Controls:
Explicit purpose-bound consent collected and logged before document submission
Disclaimer: Validations run against local mock_gstin_registry.csv and mock_udyam_registry.csv. System is not integrated with live government portals and is not DPDP-certified.

Notes:
"A major drop-off point in the baseline funnel is KYC, where 31.7% of leads abandon the process. We address this with dual-track digital verification: GSTIN for formal businesses and Udyam registration for micro-enterprises. Documents are processed locally via OpenCV and Tesseract OCR, validated against our mock registries, and stored alongside explicit user consent records."

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Fulfilment & Lifecycle

Approval-Gated Orders & Retention Tracking

1. Server-Gated Order Progression

Orders remain locked until server verification returns an approved KYC status
Tamper-proof status checks prevent direct navigation to checkout without complete compliance records

2. Token Payment Simulation

Generates a simulated ₹499 booking token payment intent
Automatically outputs a downloadable, watermarked demo PDF receipt
Clarification: Simulated payment workflow; no live banking gateways connected and no real currency debited.

3. Customer Service & Churn Prevention

Post-booking customer hub displaying simulated delivery stages and installation tickets
Integrated feedback capture routes dissatisfied ratings directly to an operational retention queue
Note: Milestone tracking is simulated and not connected to live Airtel OSS/BSS or engineer dispatch.

Notes:

"Order integrity is enforced through strict backend gates: an order cannot proceed to payment without server-verified KYC approval.
We simulate a ₹499 booking payment and generate an immediate confirmation receipt. Once onboarded, customers track deployment stages on a self-service portal, where low feedback scores automatically trigger retention workflows."

Assisted Sales

Mitra Channel Routing & Operational Control

Rule-Based Lead Recovery Engine:
Evaluates abandoned journeys using a deterministic propensity score based on cart value, completed stages, and drop-off points
Clarification: Logic is rules-based prioritization; does not use machine learning.

Four-Level Hierarchical Routing:
Level 1: Exact 6-digit Pincode match to local Mitra agent
Level 2: 3-digit Pincode prefix cluster match
Level 3: City / Circle municipal territory queue
Level 4: Central tele-sales pool escalation

Partner Safety & Central Governance:
Atomic Claiming: Concurrency locks prevent duplicate partner outreach to the same lead
Partner Scoping: Mitra agents only access customer records explicitly routed to their territory
Operations Visibility: Filterable pipeline views, drop-off analytics, and bulk CSV lead exports
Prototype Note: Ops dashboard subject to known hard-coded KPI fallback values.

Notes:
"Digital self-service is backed by our physical partner channel. Abandoned leads are scored by a rule-based engine and routed through a four-level geographical hierarchy down to the local Mitra agent. Atomic locks prevent duplicate calls, while our central operations portal maintains end-to-end visibility across the recovery pipeline."

Business Case

Pilot Impact Hypotheses vs. Baseline

Controlled Pilot Model (Per 5,000 Inbound Leads)

Pilot Assumption	Projected Delta	Pilot Assumption	Projected Delta
Activation Conversion	7.36% (368 units)	**11.00%** (550 units)	**+182 Net Activations (+49.5%)**
Mean Time to Activation	17.62 Days	**12.00 Days**	**-5.62 Days (-31.9%)**
Status-Chasing Inquiries	3.01 Calls / Customer	**2.00 Calls / Customer**	**-33.5% Support Contact Overhead**
Cost per Activation (CAC)	₹15,053.37	**₹12,200.00**	**-₹2,853.37 per customer (-18.9%)**
Post-Quote KYC Drop-off	31.71% abandonment	**22.00% abandonment**	**+9.71% Completion Recovery**

Core Underlying Drivers:

Instant digital quoting and self-serve ordering reduce sales turnaround times
Dual GST/Udyam processing recovers micro-merchants previously lost to document complexity
Mitra lead routing recovers high-value abandonments through targeted human assistance

Notes:

"These projections reflect our pilot hypotheses. By removing quoting bottlenecks and offering dual-track verification, we target lifting conversion from 7.36% to 11.00%—yielding 182 additional business customers per 5,000 leads. We project reducing average onboarding time to 12 days and lowering customer acquisition costs to ₹12,200."

Illustrative assumptions requiring controlled pilot validation. The current prototype does not measure real-world field TTA, production PBX calls, actual CAC, or 6-month churn.

Proposed Next Steps

Controlled Pilot Scope & Technical Enablers

The Implementation Ask (What We Need):

1. Airtel Systems Access: Sandbox API access to Airtel CRM, GIS feasibility checks, and OSS/BSS provisioning
2. Channel Credentials: Production Meta WhatsApp Business API sandbox environment
3. Compliance Review: Enterprise infosec and legal evaluation of client-side OCR and consent vault models
4. Pilot Inbound Cohort: Controlled allocation of 1,000 active inbound MSME leads across two target urban circles

8-Week Pilot Execution Plan:

Weeks 1–2: Sandbox API integration (CRM, BSS, live WhatsApp gateway)
Weeks 3–4: Mitra field partner onboarding and role-based access training
Weeks 5–7: Live traffic execution across 1,000 lead pilot cohort
Week 8: Pilot evaluation against baseline metrics and enterprise scaling decision

Measurable Pilot Targets:

Validate reduction in real-world Time to Activation (target: ≤ 12 days)
Verify Gemini response guard performance and zero-hallucination guardrails under production query loads
Measure active Mitra partner adoption and lead claim resolution speed

Notes:

"To prove this business case, we are asking for an 8-week controlled pilot with 1,000 inbound MSME leads. We need sandbox access to Airtel's CRM and OSS/BSS systems, along with Meta WhatsApp credentials. This pilot will validate our onboarding speed, test our AI guardrails under real-world conditions, and confirm that Project Udyam Digital is ready for nationwide deployment."